

What are GPT-based AI writing tools?

GPT (Generated Pre-Trained Transformer) is an AI-based language model trained on massive amounts of data from the internet to generate generally coherent, human-like pieces of writing on a given topic. At its core, GPT is a language model, not a knowledge model, and therefore its main contribution is to automate generation of a human-equivalent text, not to produce information. It generates text on a variety of topics in a variety of formats based on a prompt or command provided to it. It can be used for a variety of tasks, including text completion, translation, question-answering, summarization, and more. Most GPT-based tools, in particular OpenAI's ChatGPT, are designed as "chatbots" and so provide responses in a conversational format. Additionally, it is possible to correct and improve the AI's responses iteratively, essentially "teaching" the AI in the course of the interaction with the tool. OpenAI, the company that launched ChatGPT and is partnered with Microsoft, regularly releases updated versions featuring improved accuracy and performance as a result of ongoing research and development.

While ChatGPT and other GPT-based AI tools have legitimate educational uses, they also raise significant academic integrity concerns. Specifically, because GPT can produce well-composed human-like text, the text which it produces can easily be plagiarized and submitted as part of an assessment and would be very hard to detect. While there are tools which aim to detect AI-generated text (and OpenAI has released a similar tool which has since been shuttered), the sensitivity of these tools is too low to be used as reliable evidence of plagiarism, as it generates too many false positives.

Despite its ability to compose human-like text pertaining to a variety of tasks, GPT is generally not very good at critical thinking, deeper, contextual understanding, creativity, and emotional expression. Responses may lack common sense, demonstrate bias, exhibit limited domain expertise, provide false information, or create illogical texts. For a summary overview of ChatGPT, watch [this video](#) from the Office of Academic Technology.

How can students use ChatGPT and other similar AI tools

In general, GPT is very good at producing human-like text with mostly accurate and specific information across a wide range of topics. In many ways, it is best at producing "templates of speech" – representations of the general structure of different types of texts. Students can use these templates as starting points for a variety of different writing tasks, ranging from drafting emails, to summarizing information on a topic, to creating a rough draft for an entire assessment. Under this framework, the GPT-generated response serves as an initial starting point for these assignments which students can use as a base to create their own piece of writing, or, in theory, which students might submit with few to no modifications as their own work.

It is important to note that while OpenAI's ChatGPT is not connected to the internet, other GPT-based language models, such as Microsoft's Bing AI and Google Gemini, are, and can source their information from live and up-to-date internet searches. As such, students are able to use these GPT-based tools to generate written material based on specific sources from the internet, such as summaries of a specific source or sources connected to a search term.

On the other hand, because the results from GPT are intended to be as human-like as possible, students are also able to submit text from ChatGPT or other tools directly or with minimal intervention as their own work. According to the [Center for Teaching and Learning](#), papers generated by ChatGPT routinely scored B+ to A- grades on a set rubric. This behavior definitionally constitutes plagiarism, as students are submitting work which they themselves did not produce. In addition, features of other GPT-based tools can make it even easier for students to submit GPT-generated text as their own work, such as Bing AI's conversational styles.

How to address AI usage in your classroom

Address AI usage in the classroom directly. Be open and clear about your expectations about AI use: define expectations about appropriate and inappropriate use, and reinforce those expectations frequently. Design assessments bearing AI tools in mind. Inform yourself (the instructor) about GPT-based tools, familiarize yourself with them first-hand, and try feeding them your assessment prompts to find out how ChatGPT may respond. Consider the following concrete steps, many of which are suggestions from the [Center for Teaching and Learning](#):

- Determine an explicit policy for ChatGPT use in your class and include a statement about it in the syllabus. This statement can define expectations about appropriate and inappropriate use, and will set the tone for students for ChatGPT in the class. Contact your project manager if you would like assistance with creating syllabus language which aligns with your AI use goals.
- Consider requiring and enforcing strict source citation requirements for those sources used by students in constructing their responses to assessment prompts.
- Consider requiring the use and citation of specific sources, especially those sources which GPT cannot easily access, such as specific class lectures or slides or other unique information from the course, and sources only accessible through subscription services e.g. UT Libraries, such as academic papers or books which would not be indexed on normal search engines.
- Make sure to scaffold assignments so that students have enough time and space to complete each assignment. If students feel like they have the bandwidth to complete the assessment themselves, they may be less likely to turn to AI to complete it for them.
- Familiarize yourself with GPT-based tools and the kind of texts they produce. Feed some examples of assessment prompts to various tools and see what sort of responses each gives.
- Create assessments which prioritize creative or critical thinking. While GPT-based tools are very good at producing factual documents, they struggle with prompts which require deeper understanding.
- Require students to note where they used AI in their writing and how. This strategy legitimizes AI as a tool which can help them in their writing and dissuades them from using it in a way which constitutes plagiarism. (See below for more on How to incorporate GPT-based tools into your classroom.)

How to detect AI-based plagiarism

Despite an instructor's best effort, some students will still resort to plagiarizing from any of the GPT-based tools. In these cases, it can be difficult to determine whether a student's response to an assessment has been copied from such a tool, as the results are very close to human-written text. It is very important to note that preexisting tools for detecting plagiarism, such as TurnItIn, have a very hard time detecting copying from AI-generated text. In addition, there is no tool which can reliably detect whether a text is AI-generated. As such, at this time we heavily advise against using the results from such tools and relying on that result to determine whether a given response is AI-generated and therefore plagiarized.

The most effective strategy to date for assessing whether a response has been plagiarized from a GPT-based tool is to **familiarize oneself with the responses ChatGPT and other similar tools give for the prompt in an assignment so as to understand what an AI-generated response to that prompt looks like.**

- We advise creating a bank of responses for each prompt in each assessment to be shared with TAs or graders with explicit instruction to compare the language in each student's response to the sample AI-generated response.

- Given the proliferation of separate GPT-based tools, we additionally suggest including several sample responses from different tools, particularly Bing AI and Gemini, as well as several responses from those tools, such as for the different conversation styles in Bing AI.

How to incorporate AI into your classroom

Incorporate instruction on how to use AI in a constructive and educational way. One of the best ways to obviate the concerns about plagiarism from GPT-based tools is to incorporate their use in the classroom directly. If students feel as if they have agency over their work and that AI can help them in achieving their learning goals, they may be less likely to use it for plagiarism and more likely to use it for constructive and educational ends. The Center for Teaching and Learning has created [a collection of ideas](#) for how to include AI in the classroom. Some highlights include:

- Craft an assignment where you generate a ChatGPT output based on a prompt and ask your students to critique the response, indicating where it did a good job of articulating key points and what nuances it missed.¹
- Treat GPT-based tools as a tool that some students may want to use to help get started writing. For example, students who have difficulty starting writing assignments might be encouraged to generate a paragraph with ChatGPT as a stub that enables them to continue writing. As long as the student ultimately adds significant new material and thoroughly edits or ultimately eliminates the output from ChatGPT, they are producing a document that reflects their own work.

Resources

<https://ctl.utexas.edu/5-things-know-about-chatgpt>

<https://utexas.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=8a3281d3-a676-4186-b00e-af850109166d>

<https://ugs.utexas.edu/flags/teaching/artificial-intelligence-statement>

<https://platform.openai.com/docs/chatgpt-education>

¹ Note that there are privacy concerns with how ChatGPT collects data from users, so consider providing your students copies of responses from ChatGPT, rather than asking them to generate the responses themselves.